

exp 2.py

```
1 import pandas as pd
2 data = pd.DataFrame({
3     "Sky": ["Sunny", "Sunny", "Rainy", "Sunny"],
4     "AirTemp": ["Warm", "Warm", "Cold", "Warm"],
5     "Humidity": ["Normal", "High", "High", "High"],
6     "Wind": ["Strong", "Strong", "Strong", "Strong"],
7     "Water": ["Warm", "Warm", "Warm", "Cool"],
8     "Forecast": ["Same", "Same", "Change", "Change"],
9     "EnjoySport": ["Yes", "Yes", "No", "Yes"]
10 })
11 concepts = data.iloc[:, :-1].values
12 target = data.iloc[:, -1].values
13 specific = concepts[0].copy()
14 general = [ "?" for _ in range(len(specific)) ]
15 for i, h in enumerate(concepts):
16     if target[i] == "Yes":
17         for x in range(len(specific)):
18             if h[x] != specific[x]:
19                 specific[x] = "?"
20                 general[0][x] = "?"
21     else:
22         for x in range(len(specific)):
23             if h[x] != specific[x]:
24                 temp = [ "?" for _ in range(len(specific)) ]
25                 temp[x] = specific[x]
26                 general.append(temp)
27 print("Specific Hypothesis:")
28 print(specific)
29 print("\nGeneral Hypothesis:")
30 for g in general:
31     print(g)
```

Run exp 2



> /Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 2.py



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Specific Hypothesis:

['Sunny' 'Warm' '?' 'Strong' '?' '?']

General Hypothesis:

['?', '?', '?', '?', '?', '?']

['Sunny', '?', '?', '?', '?', '?']

['?', 'Warm', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', 'Same']

Process finished with exit code 0